

Azariah Boyd, MSc

azariahboyd23@gmail.com | (617) 322-4209 | <https://www.linkedin.com/in/azariahboyd/>

EDUCATION

Harvard T.H. Chan School of Public Health, Boston, MA

Master of Science in Environmental Health Epidemiology

May 2024

Stonehill College, Easton, MA

Bachelor of Science in Biology

May 2018

HONORS, SCHOLARSHIPS, AWARDS

Harvard Donald Hopkins Scholar(2022-2023), **NSF S-STEM-Scholar**(2016-2018), **Stonehill Class Leadership Award** (2016-2018)

SKILLS & CERTIFICATIONS

Technical Skills:

- **Data Analysis & Visualization:** Proficient in R Studio, STATA, Microsoft Excel, Word, PowerPoint, REDCap
- **Writing & Healthcare Communication:** Peer-reviewed manuscripts, RFPs, Grant writing, Bilingual Community Health Curriculum development, Policy Research Reports, Policy Briefs, IRB Protocol writing
- **Information Management:** Literature reviews (PubMed) and citation tools (Zotero)

Environmental Exposure Assessment:

- Refractometry with urine sample processing
- Portable X-ray fluorescence analyzers (pXRF)
- Condensation Particle Counter (CPC)

Molecular and Cellular Biology:

- Bradford protein assay, Western Blot, Fluorescent Microscopy, Microplate Reader, Polymerase Chain Reaction (PCR), Animal husbandry and safety (guppies, beetles), Aseptic techniques, Safety, and quality control

Certifications and Trainings:

- Human Subjects CITI Training (2024), HIPAA (2024), Certified Nursing Assistant Red Cross (2023), CPR (2023), Fundamentals of Medical Interpreting at UMass Medical School (2020), Remote Interpreting Training Center for Health and Impact (2020)

Spoken Languages:

- **Spanish:** Fluent, **English:** Native proficiency

PROFESSIONAL EXPERIENCE

Richard A. and Susan F. Smith Center for Outcomes Research, Beth Israel Deaconess Medical Center, Boston, MA

Joan and Marcel Zimetbaum Research Fellow

July 2024 – Present

- Selected from a highly competitive pool of candidates (acceptance rate <5%) to complete dedicated 1-year research program focused on health policy, the use of large national data sources, and casual inference and epidemiological statistical methods to rigorously evaluate the effects of local, state, and federal policies on health outcomes and equity
- Critically thinking about study design and analytic strategies in a thoughtful and nuanced way to the relationships between policy and health to generate meticulous scientific hypotheses, and convey research findings articulately to maximize their impact
- Leading health outcomes research on national changes to reproductive policy (e.g. Dobbs vs. Supreme Court) impact on women's mental health across income levels
- Spearheading studies on how to best optimize safety-net policies that aim to reduce food insecurity to improve health, including SNAP and WIC programs
- Execute advanced statistical modeling and data analysis in R Studio using econometric methodologies (difference-in-differences, instrumental variables) and epidemiologic approaches (stratified analyses) to quantify health outcome variations across demographic subgroups and policy environments
- Develop high-impact research deliverables including peer-reviewed manuscripts, conference abstracts, and academic poster, with findings presented at the largest health policy conference in the country
- Collaborate with multidisciplinary research teams of epidemiologists, biostatisticians, and clinicians to optimize study designs, validate analytical approaches, and ensure methodological rigor in compliance with NIH and institutional standards

Initium Health, Denver, CO

Grant Writer (Part-time)

March 2024 – Present

- Translate complex public health and clinical data into compelling, evidence-based narratives to support strategic decision-making for healthcare organizations and government agencies
- Secured over \$500,000 funding through competitive grant writing and Request for Proposal responses, demonstrating measurable value and impact
- Lead development of tailored proposals addressing healthcare access, quality improvement, and systems transformation—

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aligning each submission with funder priorities and organizational capacity

- Collaborate with multidisciplinary teams to synthesize operational data, service models, and population health indicators into persuasive content
- Ensure submissions meet compliance standards, deadlines, and formatting requirements while maintaining clarity and analytical rigor

Environmental Reproductive Justice Lab, Harvard TH Chan School of Public Health, Boston, MA

Bilingual Graduate Research Associate

Sept 2021 – May 2024

- Conducted a case-control clinical research analysis (n=30) investigating associations between endocrine-disrupting chemical exposures (e.g., phthalates, metals mixtures) and Central Centrifugal Cicatricial Alopecia among Black women, performing multivariable logistic regression analyses and producing comprehensive final report with symposium presentation
- Led environmental justice community events for pregnant people and those with young children about environmental toxins and health
- Led comprehensive exposure epidemiologic analyses on EPA standard levels of toxic metal exposures across 20 public drinking fountains to identify neighborhood proximity to lead service lines in Boston, MA, managing end-to-end sample collection logistics, quality control protocols, and database management for 40+ samples
- Crafted data visualizations and delivered evidence-based policy recommendations to academic researchers and community stakeholders on environmental exposure assessment findings at key Superfund remediation sites
- Applied the EPIS (Exploration, Preparation, Implementation, Sustainment) implementation science framework to develop strategic community engagement protocols, evaluating youth community science program effectiveness and creating systematic evaluation metrics
- Co-designed bilingual environmental justice curriculum on climate change, metals, and health reaching 100+ K-12 students over 2 years, integrating culturally responsive pedagogy with scientific concepts to promote environmental health literacy in underserved communities

Health Reform and Community Medicine, Santiago, Chile

Graduate Student Researcher

Winter Term 2023

- Conducted field interview key health system actors and coordinated site visits to clinics and hospitals to examine the political and historical context of protest movements, Constitutional reform, and health policy under the Boric administration
- Analyzed Chile's dual public-private insurance model, public-private partnerships for delivery in primary care, and family medicine integration challenges, alongside indigenous health policy developments
- Collaborated with local academics and Chilean family medicine practitioners to design an independent research study analyzing associations between air pollution and non-communicable diseases across income-stratified communities, with a focus on environmental justice and translational policy recommendations

Harvard Cyprus Endowment Program for the Environment and Public Health, Boston, MA

June 2022 – June 2023

Research Trainee

- Selected from competitive pool of applicants to complete 1-year research intensive in environmental health epidemiology, applying advanced regression techniques for interventional randomized clinical trial in Nicosia, Cyprus
- Conducted an evaluation of PM 2.5 indoor home exposure to examine the diurnal and monthly variability of particulate matter to assess the effectiveness of exposure-reduction intervention among elderly adults with atrial fibrillation in R Studio
- Prepared manuscript and presented research findings to UNIC Medical School stakeholders and research team

Massachusetts General Hospital, Disparities Research Unit, Boston, MA

Bilingual Care Manager, Clinical Research Coordinator I & II

July 2019 – Aug 2021

- Conducted weekly quantitative analyses of participant dosage and engagement data using Excel, generating actionable insights that informed clinical decision-making for pilot and full-scale study arms across 200+ participants
- Supported IRB submissions, protocol amendments, and regulatory documentation while maintaining data accuracy in REDCap for a federally funded youth suicide prevention study
- Provided bilingual care management services to Spanish-speaking participants, facilitating access to integrated healthcare services and achieving exceptional >80% study retention rate in a high-risk population
- Performed standardized mental health safety assessments and maintained comprehensive resource databases, ensuring rapid response protocols for participants experiencing crisis situations
- Supervised and mentored 8 community health workers, implementing performance optimization strategies that improved project alignment and enhanced team productivity by 25%
- Promoted from CRC I to CRC II within 12 months based on exceptional performance, taking on expanded responsibilities in protocol development and regulatory compliance

Patronato Benéfico Oriental, Inc., Stonehill Service Corp, Hogar del Niño, Dominican Republic

Aug 2018 – Aug 2019

Youth Education, Health Literacy, and Community Engagement

- Selected from a competitive pool of applicants for the Stonehill Service Corp

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- Developed engaging curricula for English, Health Biology, and Dance for students in grades 4 – 8, improving comprehension and participation
- Strengthened curriculum by incorporating interactive learning methods, including hands-on experiments and movement-based activities
- Collaborated with fellow educators to organize community-centered events, workshops, and cultural exchange programs, fostering a dynamic learning environment
- Engage with both university and foundation professional development opportunities related to community service

Stonehill College Department of Biology, Easton, MA

Sept 2015 – Aug 2018

Lab Manager, Teaching Assistant & NSF Undergraduate Researcher

- Managed daily laboratory operations, enforced safety protocols, and created a high-quality learning environment for students.
- Trained and mentored peers in laboratory procedures, scientific methodology, and data analysis techniques
- Led and supported behavioral neuroscience research on anxiety in guppies treated with Prozac; managed data collection, scheduling, and contributed to analysis
- Field work collecting Trinidadian native *Poecilia reticulata* (guppies)
- Conducted honors thesis on *Phenoloxidase (PO) Activity in Burying Beetles*: designed field and lab-based experiments including beetle breeding, secretion sampling, protein analysis via Bradford Assay, and data interpretation
- Collaborated on multiple research projects including:
 - “*Lateral line morphology influences social learning in guppies*” – collected and analyzed data using fluorescent microscopy and ImageJ; presented at the Stonehill Undergraduate Research Experience (SURE) program event
 - “*Influence of Social Environment on Guppies*” – executed social learning trials and prepared guppy biological media for gene expression analysis via PCR

PUBLICATIONS

1. Alegria, M., O'Malley, I., Smith, R., Williams, Useche-Rosania, A., **Boyd, A.**, Cuervo-Torello, F., Williams R. D., Acevedo-Garcia, D., (2022) Addressing health inequities for children in immigrant families: Psychologists as leaders and links across systems. *American Psychological Association*.
2. Liu, M., Inoue, K., **Boyd, A.**, Aggarwal, R., Marinacci., Wadhera, R. Inequities in Obesity Prevalence by Race and Ethnicity Among US Children and Adolescents, 2011-2023. *Annals*.

In progress

3. **Boyd, A.**, Liu, M., Zera C., Stevens, J., Wadhera, R. Changes in Mental Health Among Women After the Dobbs Abortion Decision
4. **Boyd, A.**, Michanikou, A., Koutrakis, P., Panayiotis K., Achilleos, S., Papatheodorou, S., Yiallourous, P., (2023) Assessing Effectiveness of Indoor Air Purification Intervention on PM Levels: The Impact of Season and Time of Day

PUBLIC SPEAKING & PRESENTATIONS

Moderator, Panel on Climate, Occupation, & Community Health, Harvard TH Chan School of Public Health	Jan 2025
Presenter, MEMCARE Superfund Research Center, Harvard TH Chan School of Public Health	March 2024
Panelist, Speaker series: Women in STEM. Women's History Month, Boston Latin School	March 2024
Speaker/Moderator, Inaugural MLK Commemorative Lecture, Harvard University	Oct 2022
Panelist, Careers in Public Health, Dorchester Boys & Girls Club	June 2022
Alumni Speaker, Health Science Course – Exploring Career Opportunities, Stonehill College	Sept 2020

COMMUNITY LEADERSHIP & COMMUNITY ENGAGEMENT

<i>BRACES Alumni Mentor</i> Stonehill College	Jan 2024 – Present
<i>Co-Chair Equity, Diversity, Inclusion & Belonging Committee</i> HSPH Dept. of Environmental Health	Nov 2024 – Present
<i>Community Outreach, BGV Restore</i> Pentecostal Tabernacle , Cambridge, MA	March 2024 – Present
<i>Co-Chair, Media and Marketing Specialist</i> Harvard Chan Africa Health Conference	Sept 2022 – Sept 2023
<i>Group Leader</i> FACETS Harvard T.H. Chan School of Public Health	June 2023 – July 2023
<i>Community Engagement Coordinator</i> Harvard Chan Student Association	Sept 2022 – May 2023

CONFERENCES AND SYMPOSIUMS

- Academy Health Meeting 2025 2025
- New England of Science Symposium, Harvard Medical School 2024
- NIH Superfund Research Center 35th Annual Meeting 2023
- Summit for Women in STEM. Wheaton College, Massachusetts 2018
- The 54th Annual Conference of the Animal Behavior Society 2017